

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 2

Year: 2024

Subject: 030523700 Selected Topics in Computer

Section: 1-2

Date: 22 January 2025

Time 15.00 - 17.00 p.m.

Examination Setter Name: Dr. Phollakrit Wongsantisuk

Student Name.....ID.....Class.....No.

Directions:

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination consists of 9 questions on 7 pages (including the cover page). The total score is 25 points.
3. Complete all questions in this test paper.
4. Books, documents and lecture notes are not allowed but dictionary are allowed.
5. Electronic calculators not allowed (if allowed; specify the models designated by the faculty).
6. Students must remain in the examination room for no less than 1 hour from the start of the examination
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency.
9. Students are not allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.
10. Usage of all communication devices during the examination is strictly prohibited.

Cheating in examinations is considered a serious offense and carries the penalty of expulsion from student status.

ข้อสอบชุดนี้เป็นส่วนหนึ่งที่ใช้ประเมินความสามารถนักศึกษา
ดังต่อไปนี้

		ข้อ1	ข้อ2	ข้อ3	ข้อ4	ข้อ5	ข้อ6	ข้อ7	ข้อ8	ข้อ9
CLO 1	ผู้เรียนสามารถเข้าใจและอธิบายความรู้วิชาการใหม่ๆ ทางด้านเทคโนโลยีคอมพิวเตอร์ได้	✓	✓							
CLO 2	ผู้เรียนสามารถใช้โปรแกรมและเครื่องมือในการแก้ โจทย์ทางด้านเทคโนโลยีคอมพิวเตอร์ได้		✓	✓	✓	✓	✓	✓	✓	✓
CLO 3	ผู้เรียนสามารถเขียนโปรแกรมหรือสร้างฮาร์ดแวร์ทาง อิเล็กทรอนิกส์และคอมพิวเตอร์เพื่อแก้ปัญหาทางด้าน เทคโนโลยีคอมพิวเตอร์ได้				✓	✓	✓	✓	✓	✓

1. Explain the five stages of the evolution of Version Control and describe the advantages and disadvantages of each stage. (3 points)

2. Explain how the commands *git pull* and *git clone* are used in different scenarios. (2 points)

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3. From the following two lines of code, explain why the error occurs and correct the code according to the principles of React JSX. (2 points)

```
const t1 = <hr/><div>Hello World</div>
const t2 = React and React Native
```

4. Complete the following JSX attribute writing correctly. (3 points)

HTML attributes	JSX attributes
class	
for	
maxlength	
tabindex	
onreset	
onmouseover	

5. Complete the following inline CSS code used with JSX correctly. (2 points)

```
const divStyle = {
  (5.1)..... // text-align: left;
  (5.2)..... // background-color: powderblue;
  (5.3)..... // font-family: verdana;
  (5.4)..... // color: red;
}
return <div style={divStyle}>Hello React</div>
```

6. When we create a Class Component and define an event handler as a regular function, we may encounter an issue where this cannot be used to refer to other methods or properties within the class. To resolve this, we can add a constructor to the class and then call the event handler method followed by `.bind(this)`, as shown in the following code.

```
import React, {Component} from 'react'
export default class Button extends Component {
  constructor() {
    super()
    this.onClickButtonOK = this.onClickButtonOK.bind(this)
  }
  showAlert(msg) {
    alert(msg)
  }
  onClickButtonOK() {
    this.showAlert('Hello')
  }
  render() {
    return <button onClick={this.onClickButtonOK}>OK</button>
  }
}
```

From the code above, if we don't want to add a constructor to the class, how can we fix it? Please modify the code so that it works without adding a constructor to the class. (3 points)

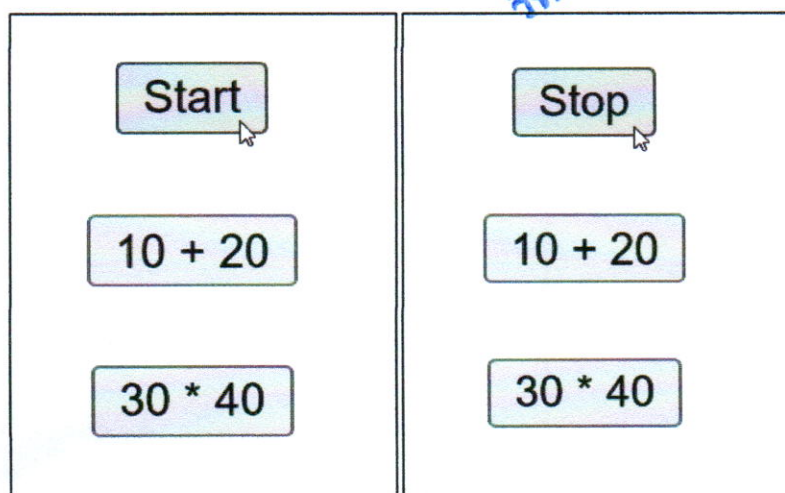
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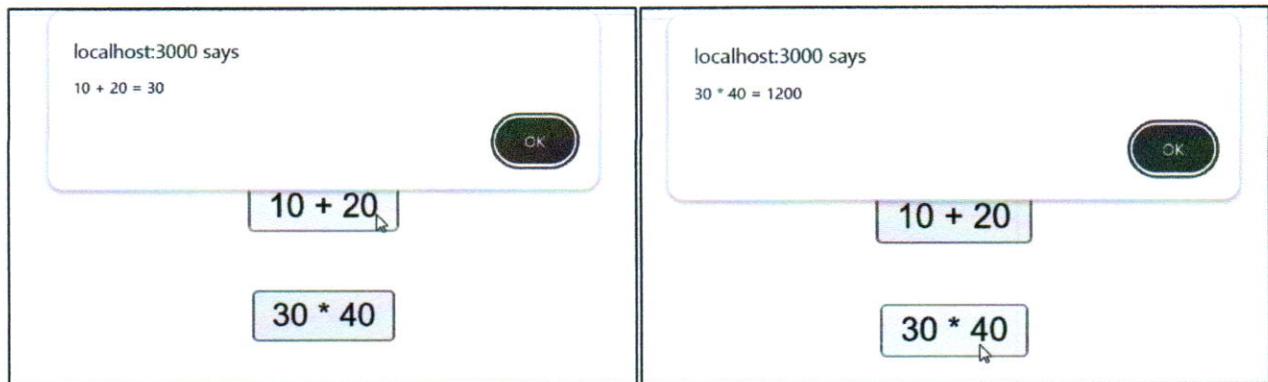
7. Fill in the blanks in the event-data.js file to complete it, so the program works as shown in the image below. (4 points)

event-data.js

```
import (7.1)..... from 'react'

export function EventData2() {
  const (7.2)..... = (ev) => {
    let t = ev.target.innerText
    let (7.3).....
    ev.target.innerText = s
  }
  const onClickAdd = (ev) => {
    let t = ev.target.innerText
    let r = eval(t)
    alert((7.4).....)
  }
  return (
    <div style={{textAlign:'center', marginTop:20}}>
      <button onClick={(ev) => onClickStartStop(ev)}>(7.5).....</button><br/><br/>
      <button onClick={(7.6).....}>
        (7.7).....</button><br/><br/>
      <button onClick={(7.8).....}>30 * 40</button><br/><br/>
    </div>
  )
}
```





8. Explain the reasons why, in React, it is recommended to create element references (Refs or References) and access elements through these references instead of referring to elements using JavaScript directly. Also, complete the missing parts of the example program for creating a reference in a Class Component. (3 points)

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```
import React, { (8.1)..... } from 'react'
class MyClass extends React.Component {
  buttonOK = createRef()
  ...
}
...
return <button (8.2).....>OK</button>
```

9. Explain the meaning of the missing parts in the following example of setting state for a Function Component. (3 points)

```
import React, {useState} from 'react'

function MyComponent() {
  let [fontSize, setFontSize] = useState(16)
  ...
}
```

- [fontSize, setFontSize] is the creation of variables using array destructuring.
- fontSize is (9.1).....
- setFontSize is (9.2).....
- useState(16) is (9.3).....

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